22. The inverse of $f(x) = \begin{cases} x & \text{if} \quad x < 1 \\ x^2 & \text{if} \quad 1 \le x \le 4 \\ 8\sqrt{x} & \text{if} \quad x > 4 \end{cases}$ is

(1) $f^{-1}(x) = \begin{cases} x & \text{if} \quad x < 1 \\ \sqrt{x} & \text{if} \quad 1 \le x \le 16 \\ \frac{x^2}{64} & \text{if} \quad x > 16 \end{cases}$

(2) $f^{-1}(x) = \begin{cases} -x & \text{if} \quad x < 1 \\ \sqrt{x} & \text{if} \quad 1 \le x \le 16 \\ \frac{x^2}{64} & \text{if} \quad x > 16 \end{cases}$

(3) $f^{-1}(x) = \begin{cases} x^2 & \text{if} \quad x < 1 \\ \sqrt{x} & \text{if} \quad 1 \le x \le 16 \\ \frac{x^2}{64} & \text{if} \quad x > 16 \end{cases}$

(4) $f^{-1}(x) = \begin{cases} 2x & \text{if} \quad x < 1 \\ \sqrt{x} & \text{if} \quad 1 \le x \le 16 \\ \frac{x^2}{8} & \text{if} \quad x > 16 \end{cases}$

23. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 1 - |x|$. Then the range of f is

(1) $\mathbb{R}$ (2) $(1, \infty)$ (3) $(-1, \infty)$ (4) $(-\infty, 1]$

24. The function $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = \sin x + \cos x$ is

(1) an odd function
(2) neither an odd function nor an even function
(3) an even function
(4) both odd function and even function.

25. The function $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \frac{(x^2 + \cos x)(1 + x^4)}{(x - \sin x)(2x - x^3)} + e^{-|x|}$$

is (1) an odd function (2) neither an odd function nor an even function
(3) an even function (4) both odd function and even function.

Summary

In this chapter we have acquired the knowledge of

- Set
  - Subset, super set, trivial subset, proper subset, improper subset
  - Empty set, power set, universal set, singleton set, finite set, infinite set
  - Cardinality of a set
  - Union, Intersection, Complement, Set Difference, Symmetric Difference
  - Properties and De Morgan Laws
  - Cartesian Product
- Intervals
  - Constants, dependent and independent variables
  - Open, Closed, finite and infinite intervals and neighbourhoods;